Chapter 8. Baire Spaces and Dimension Theory

The Baire category theorem, nowhere-differentiable functions, covering dimension and embeddings

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8.1 Baire spaces

A set A has **empty interior** if it contains no nonempty open set. X is a **Baire space** if every countable union of closed sets with empty interior has empty interior; equivalently, every countable intersection of dense open sets is dense.

Theorem 8.1 (Baire category theorem)

Every compact Hausdorff space and every complete metric space is a Baire space.

Proof. Let A_n be closed with empty interior and U_0 nonempty open; we find a point of U_0 outside $\bigcup A_n$. Since A_1 has empty interior, $U_0 \not\subset A_1$; choose a nonempty open U_1 with $\overline{U_1} \subset U_0 - A_1$ (regularity), and in the metric case with diameter < 1. Continue: $\overline{U_n} \subset U_{n-1} - A_n$, diameter $< 1/n$. The nested nonempty closed sets $\overline{U_n}$ have a common point (by compactness, or because centres form a Cauchy sequence), which lies in U_0 and in no A_n. ■

Open subspaces of Baire spaces, and locally compact Hausdorff spaces, are Baire. $\mathbb{Q}$ is not Baire (a countable union of points).

Theorem 8.2

If X is Baire and $f_n : X \to Y$ (Y metric) are continuous with $f_n \to f$ pointwise, then the set of points where f is continuous is dense in X.

Idea. For $\epsilon > 0$ the closed sets $A_N = \{x : d(f_n(x), f_m(x)) \le \epsilon \ \forall n, m \ge N\}$ cover X, so the union of their interiors is dense (Baire); on the interior of A_N, f is within 3ϵ of the continuous f_N locally, so its oscillation is at most 3ϵ. Intersect over $\epsilon = 1/k$. ■

For example, derivatives (pointwise limits of difference quotients) are continuous on a dense set; the indicator of $\mathbb{Q}$ is not a pointwise limit of continuous functions.

8.2 A nowhere-differentiable function

Theorem 8.3

In the complete metric space $\mathcal{C}([0,1], \mathbb{R})$ with the sup metric, the functions that are differentiable at some point form a set of first category (a countable union of closed sets with empty interior). In particular continuous nowhere-differentiable functions exist and are dense.

Idea. For n, let F_n be the set of f for which some x has $|f(x+h) - f(x)| \le n|h|$ for all small h (say $0 < |h| \le 1/n$, within $[0,1]$). F_n is closed (compactness of $[0,1]$). It has empty interior: every continuous f is uniformly close to a piecewise linear function, and adding a tiny but very steep sawtooth gives a function outside F_n. Every function differentiable somewhere lies in some F_n; by Baire, their union is not everything. ■

8.3 Introduction to dimension theory

A collection of sets has **order** $m+1$ if some point lies in $m+1$ of its members but none lies in more. A space X has **(covering) dimension** at most m if every open cover has an open refinement of order at most $m+1$ that covers X; $\dim X$ is the least such m.

Theorem 8.4

(a) A closed subspace of X has dimension at most $\dim X$. (b) If $X = Y \cup Z$ with Y, Z closed of finite dimension, $\dim X = \max(\dim Y, \dim Z)$. (c) $[0,1]$ has dimension 1; every compact subspace of $\mathbb{R}^m$ has dimension at most m.

For (c). Cover $\mathbb{R}^m$ by small cubes and perturb their positions so that at most $m+1$ cubes of a suitable family meet at any point (a "brick" arrangement); by a Lebesgue number argument, small enough bricks refine any open cover of a compact set. ■

Points of $\mathbb{R}^N$ are in **general position** if any $k+1 \leq N+1$ of them are affinely independent; general position can be achieved by arbitrarily small perturbations.

Theorem 8.5 (Imbedding theorem)

Every compact metrizable space of dimension m imbeds in $\mathbb{R}^{2m+1}$. In particular every compact m-manifold imbeds in $\mathbb{R}^{2m+1}$.

Idea. In the complete metric space $\mathcal{C}(X, \mathbb{R}^{2m+1})$, the maps that are "$\epsilon$-injective" (points with the same image are within ϵ) form a dense open set: given f, take a fine open cover of order at most $m+1$, a partition of unity ϕ_i, and points z_i in general position near f of the cover sets; $g = \sum \phi_i z_i$ is close to f, and general position (two m-simplices in $\mathbb{R}^{2m+1}$ in general position are disjoint unless they share vertices) makes g ϵ-injective. By Baire, the intersection over $\epsilon = 1/n$ is nonempty, giving an injective map, hence an imbedding (compact domain). ■

Compact m-manifolds have dimension m, and $2m+1$ is sharp in general (some graphs do not imbed in $\mathbb{R}^2$, Chapter 10).